

STATEMENT OF WORK: LN2-Cooled Mid-Wave Camera Acquisition and Delivery

NASA Contract 80NSSCxxxxx

**National Aeronautics and Space Administration (NASA)
Ames Research Center
Moffett Field, California, 94035**

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Statement of Work: LN2MW Camera Acquisition

NASA Contract 80NSSCxxxxxxx

1 SCOPE

This Statement of Work (SOW) describes the contracted acquisition of a Liquid Nitrogen Cooled Mid-Wave (LN2MW) Camera to NASA. This acquisition is derived from a previous contract awarded to IRCameras, LLC under NASA Order Number 80NSSC24PC263 involving the design and development of the LN2MW Camera based on the IRCameras Model IRC806.

2 OBJECTIVES

NASA is acquiring the LN2MW Camera to conduct laboratory and field measurements using thermal and radiometric imaging methods at infrared (IR) wavelengths. The LN2MW Camera will complement and augment NASA's experimental facilities and capabilities that support the Agency's applied research missions.

3 TECHNICAL APPROACH

3.1 General

The LN2MW Camera acquisition involves the procurement, fabrication, assembly, qualification-testing, documentation, and delivery of the LN2MW Camera and associated material to NASA. The contracted work shall adhere to technical specifications included in Attachment 1. The camera LN2MW Camera shall incorporate the mechanical, optical and electronic features identified by the detailed design described in References 1 and 2.

3.2 Procurement, Fabrication, Assembly

3.2.1 Contractor Furnished Camera Components

The Contractor shall procure the necessary materials, equipment and services required to construct, assemble and qualify the LN2MW Camera. Affected camera assemblies and components include the items described in Reference 1:

3.2.1.1 IRC806 upward looking camera assembly featuring the LN2-cooled focal plane array (FPA), LN2 dewar reservoir, LN2 fittings, long-pass cold filter, and filter wheel assembly

3.2.1.2 Integrated electronics interface box

3.2.1.3 Power adapter with right-angle jack plug connector to camera

3.2.1.4 Removable bayonet type f2.3 lens

3.2.2 NASA Furnished Camera Components

NASA shall provide up to four optical filters for the Contractor to install into the camera filter wheel prior to the LN2MW Camera's final assembly.

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3.3 Qualification Testing

Following camera manufacture and assembly, the contractor shall perform the necessary qualification testing of the LN2MW Camera. These tests shall generally include, but are not necessarily limited to:

3.3.1 Vacuum leak testing of the LN2 dewar.

3.3.2 Measuring and verifying predicted LN2 hold times during camera operation.

3.3.3 Demonstrating the camera achieves the operational and technical performance requirements specified in Attachment 2 of this SOW.

3.4 Documentation

The vendor shall furnish all required documentation necessary to configure, operate, and maintain the LN2MW Camera. The documentation shall include, but not necessarily be limited to:

3.4.1 Full description of software commands and data formats required for camera set-up, communication, control, and image acquisition from a host computer.

3.4.2 A detailed description of the 14-bit digital video and metadata stream transmitted to the host computer through the Camera Link and Gigabit Ethernet interfaces.

3.4.3 An electrical interface description document (IDD) describing the signals available at the camera's electrical, video and timing interfaces.

3.4.4 Two-dimensional line drawings, CAD files, other schematics showing the camera external dimensions, mechanical features and fastening interfaces

3.4.5 Certificate of volatility documenting the applicable presence or absence of data storage memory and media onboard the camera (refer to sample certificate furnished in Appendix B).

3.4.6 Acceptance test report with tabulated or graphical data including, but not necessarily limited to, the following information:

3.4.6.1 FPA pixel noise histogram (raw and NUC-adjusted pixels).

3.4.6.2 Measured NEDT.

3.4.6.3 Map of inoperative FPA pixels.

3.4.7 Spectral response of FPA detector.

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4 BRIEFINGS, REVIEWS

The Contractor shall provide the designated NASA technical task manager with timely communication of technical progress and promptly report any issues that may impact the successful execution and completion of SOW task elements. The Contractor is encouraged to use informal means (including email, telephone and internet conference calls) to keep the NASA technical task manager apprised of the work being performed under this contract.

5 GOVERNMENT FURNISHED EQUIPMENT (GFE)

NASA shall deliver to the Contractor up to four (4) optical filters as Government Furnished Equipment (GFE). Details regarding the method and schedule of delivery of these GFE items shall be mutually agreed upon by NASA and the Contractor no later than four (4) weeks following contract authorization.

6 DELIVERABLES

The Contractor shall deliver all required LN2MW Camera equipment and materials specified in this SOW to the shipping address identified in the NASA contract.

The Contractor shall provide NASA with an electronic copy of presentations, design documents, drawings, reports, parts lists, technical specifications and other relevant material generated under this contract.

The Contractor shall provide narrative reports in electronic form in either Adobe PDF or Microsoft Office Word file format. Mechanical line drawings shall be furnished in DXF, DWG, or Adobe PDF file format. Computer aided design (CAD) files shall be provided in either STP or STEP format.

7 WARRANTY

All hardware and software shall be warranted to be free of defects. The warranty shall cover all parts and labor for a period of one-year following LN2MW Camera delivery to NASA.

8 DATA AND PROPERTY RIGHTS

The following data and property rights apply to the work performed by the Contractor under this contract:

- 8.1 NASA shall retain ownership of all purchased hardware.
- 8.2 NASA shall have unrestricted data rights on all Contractor generated data.
- 8.3 Contractor proprietary information used to support this task shall remain proprietary to the Contractor, unless otherwise agreed upon by both NASA and Contractor.
- 8.4 The Contractor shall clearly identify in their proposal any information that it considers to be Contractor-proprietary technology.

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- 8.5 Where appropriate, Contractor proprietary information shall be submitted in documents or files that are separate from non-proprietary information that is furnished to NASA as part of a contract deliverable.

9 TRAINING

The Contractor shall provide NASA personnel with hands-on demonstration and training on LN2MW Camera set-up, operation and maintenance. Details regarding the training arrangements shall be mutually agreed upon by NASA and the Contractor during the performance of this contract.

10 TRAVEL

If Contractor deems that travel to NASA is essential in the performance of this contract after initial contract award and authorization, the Contractor shall obtain the necessary approvals from the NASA contracting officer before commencing travel.

11 PERIOD OF PERFORMANCE

NASA's estimated period of performance for the Contractor to complete the work under this contract is six (6) months following contract authorization to proceed.

12 REFERENCES

- 12.1 Final Report for Contract 93458: NASA Upward LN2 Design, IRCameras, LLC Document 319-004-059 (Proprietary), NASA Order Number 80NSSC24PC263

- 12.2 Electronic file name: 319-004-028_X3.STEP

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Attachment 1

LN2MW Camera Technical Specifications

LN2 Cold Mid-Wave (LN2MW) Camera

Technical Specifications

1 Introduction

NASA has a requirement for a liquid nitrogen (LN2) cooled mid-wave (LN2MW) camera. The LN2MW Camera will complement existing camera models owned and operated by NASA for conducting laboratory and field measurements involving thermal and radiometric imaging at infrared (IR) wavelengths.

The LN2MW Camera under procurement shall be operationally interchangeable with NASA's existing Model IRC900-series cameras to facilitate rapid equipment integration and operation with facility and test instrumentation. The LN2MW Camera will be subjected to a range of environmental and imaging extremes similar to those subjected on the IRC900 camera units.

Technical specifications for camera features, operation and measurement performance are described in the following section.

2 Specifications

The LN2MW Camera shall meet or exceed the specifications described in the following subsections.

2.1 Detector Configuration and Performance

- 2.1.1 Focal plane array (FPA) detector type: Indium Antimonide (InSb).
- 2.1.2 FPA resolution: 640 horizontal X 512 vertical pixels.
- 2.1.3 Pixel pitch size: 20 μm to 25 μm .
- 2.1.4 FPA detector response: <3 to 5.3 μm
- 2.1.5 FPA Sensitivity: NEDT < 20 mK at +30°C.
- 2.1.6 Detector well capacity: $\geq 7 \times 10^6$ electrons.
- 2.1.7 14-bit (or higher) dynamic range
- 2.1.8 FPA detector cooling by pour fill LN2 dewar for ≥ 4 hour hold time and camera operation.
- 2.1.9 FPA pixel operability.
 - 2.1.9.1 Definition

A pixel is considered operative if the following conditions are met following a two-point non-uniformity correction (NUC) process using a blackbody source set to temperatures of +20°C and +50°C:

- 2.1.9.1.1 The camera integration time is selected to provide a nominal 80% FPA well fill with the camera observing a +60°C blackbody source at a 60 Hz frame rate with no lens or warm filters installed.
- 2.1.9.1.2 The calculated NUC pixel gain value ranges between 0.8 and 1.2.
- 2.1.9.1.3 The calculated NUC pixel offset value ranges between -8,191 and +8,191.
- 2.1.9.1.4 The NUC-adjusted pixel exhibits no flickering behavior while observing the blackbody at either blackbody NUC temperature settings during a continuous 1000 frame observation period (minimum).
- 2.1.9.2 Distribution of non-operative pixels in the FPA.
 - 2.1.9.2.1 99.8% for central square region of the FPA containing at least 25% of the total pixels.
 - 2.1.9.2.2 99.5% for remaining pixels outside of the central square region of the FPA.
- 2.1.9.3 Non-operative pixel clusters.

Non-operative (bad) pixel clusters in the FPA are subject to the following constraints based on the FPA regions defined in Figure 1:

C1 160 X 128	C2 320 X 128	C3 160 X 128
C4 160 X 256	B 320 X 256 A 80 X 64	C5 160 X 256
C6 160 X 128	C7 320 X 128	C8 320 X 128

Figure 1 – FPA Regions

- 2.1.9.3.1 No clusters of 5x5 bad pixels are allowed in zones A, B or C unless within 5 pixels of an edge.
- 2.1.9.3.2 No clusters of 3x3 bad pixels are allowed in zone A.

- 2.1.9.3.3 ≤ 2 clusters of 3x3 bad pixels are allowed in zone B exclusive of zone A.
- 2.1.9.3.4 Total of ≤ 3 clusters of 3x3 bad pixels are allowed in zones C2, C4, C5 and C7.
- 2.1.9.3.5 Total of ≤ 6 clusters of 3x3 bad pixels are allowed in zones C1, C3, C6 and C8.

2.2 Optical Components

2.2.1 Cold shield and field stop sized for F/2.3 lenses.

2.2.1.1 No FPA corner pixel obscuration or vignetting by the lens barrel or filter aperture.

2.2.1.2 The FPA and lens entrance aperture shall yield optical scatter sensitivity values (S, as defined in Appendix A) no less than 0.97 under the following conditions:

2.2.1.2.1 A 50 mm (or shorter focal length lens) is attached to the camera.

2.2.1.2.2 The warm filter is absent from the camera's optical train.

2.2.2 Custom long-pass cold (LN2 cooled) IR filter having the following on-axis transmission characteristics:

2.2.2.1 λ_1 at 80% of first peak transmission: $3.0 \mu\text{m} \pm 0.1 \mu\text{m}$ (refer to Figure 2).

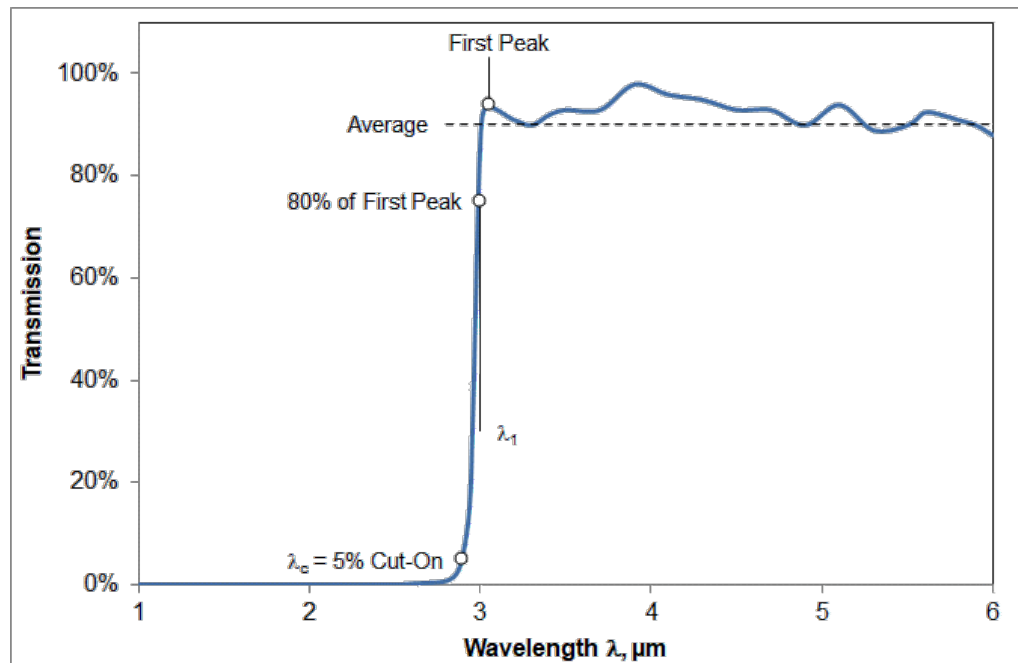


Figure 2 – Long-pass cold filter performance parameters

- 2.2.2.2 Average transmission from 3.2 to 5 μm : $\geq 90\%$.
- 2.2.2.3 Minimum transmission from 3.2 to 5 μm : $\geq 80\%$.
- 2.2.2.4 Average transmission from 1.5 to 2.9 μm : $< 0.1\%$.
- 2.2.2.5 Slope, $(\lambda_1 - \lambda_c)/\lambda_c$: $< 2.7\%$.
- 2.2.3 LN2 cooled motorized filter wheel to accommodate up to four (4) 25.4 mm diameter X 0.5 to 1.5 mm thick optical filters.
- 2.2.4 OPTIONAL: customer- or camera manufacturer-furnished optical filters for installation into LN2 cooled motorized filter wheel by camera manufacturer.
 - 2.2.4.1 Up to four (4) optical filters.
 - 2.2.4.2 For camera manufacturer-furnished filters, the customer will specify the spectral transmission performance of each filter.
- 2.2.5 Lens
 - 2.2.5.1 F/2.3 lenses with the features shown in Table 1:

Focal Length	Field of View	Focusing Method	Quantity	Accessory
25 mm	37°	Manual	1	Optional
50 mm	18°	Manual	1	Required
100 mm	9°	Manual	1	Optional

Table 1 – Camera Lens Configurations

- 2.2.5.2 Mechanical twist-lock bayonet for lens attachment to camera housing.
- 2.3 On-board Electronics
 - 2.3.1 Integration time adjustment:
 - 2.3.1.1 Adjustable up to the full frame rate.
 - 2.3.1.2 Multiple integration (super-framing) image acquisition mode.
 - 2.3.1.3 Up to four separate, user-defined integration (exposure) times during super-framing camera operation.
 - 2.3.2 Adjustable frame rate to above 100 frames/sec (full frame).
 - 2.3.3 Temperature sensor locations
 - 2.3.3.1 FPA detector.
 - 2.3.3.2 Camera internal housing.
 - 2.3.3.3 LN2 cooled filter wheel (optional).

2.3.4 Image transmission interfaces

2.3.4.1 Live 14-bit (or higher) digital image transmission over Camera Link and Gigabit Ethernet interfaces.

2.3.4.1.1 Raw (non-NUC adjusted) image frames.

2.3.4.1.2 Frame metadata embedded with each image frame including, but not limited to, the example parameters identified in Table 2 (listed in no particular order or preference):

Parameter	Comments, Description
Time	Camera IRIG timestamp (μ sec resolution)
FPA Temperature	Temperature of FPA detector, degrees Kelvin, 0.1 deg resolution or higher
Board Temperature	Temperature of FPGA electronics internal to camera, degrees Celsius, 0.1 deg resolution or higher
Integration Time	In μ sec
Filter Position	Values of 1 to 4
Frame Rate	Full Frame, In Hz
Frame Size	Height, width (in pixels)

Table 2 – Example Frame Metadata Parameters

2.3.4.2 Live video output stream of user-selected raw or NUC-adjusted camera imagery in either HDMI or SDI format.

2.3.5 Video, Communication Interfaces

2.3.5.1 Serial communication for camera control from host computer via Camera Link and Gigabit Ethernet interfaces.

2.3.5.2 14-bit (or higher) digital video transmission to a host computer via the Camera Link interface up to the full frame rate.

2.3.5.3 Coaxial jack receptacle for frame synchronization, clock-in.

2.3.5.4 Coaxial jack receptacle for frame synchronization, clock-out.

2.3.5.5 Coaxial jack receptacle for HDMI or SDI video out.

2.3.6 Electronic Memory Usage

2.3.6.1 No internal storage media (magnetic or solid-state), or portable non-volatile (NV) memory (such as SD cards) shall be used for retaining raw or NUC-adjusted FPA images within the camera unit.

2.3.6.2 Any in-situ, non-uniformity correction (NUC) data generated from live camera imagery shall be stored on either volatile memory (NUC data erased from volatile memory upon camera power-down), or portable NV memory (such as SD cards). The NUC data includes pixel offset information that is typically created by the user during the one-point NUC or offset process.

2.3.6.3 Camera configuration settings shall be stored on either embedded or portable NV memory (such as SD cards); NV memory containing user-modifiable camera configuration information must be capable of being reset or restored to the factory default configuration by the user.

2.3.7 Wireless Prohibition

The camera shall contain no onboard Wi-Fi, Bluetooth, infrared, or other form of wireless data transmission or wireless networking capabilities.

2.3.8 Input power

2.3.8.1 Twist-lock type bayonet receptacle for 18-30 VDC input.

2.3.8.2 AC-DC adapter for camera connection to 115 VAC/60 Hz/1-Ph source.

2.4 Physical Features

2.4.1 Pour Fill LN2 Dewar

2.4.1.1 The LN2MW Camera assembly shall incorporate a pour-fill LN2 dewar for cooling both the FPA and filter wheel assembly.

2.4.1.2 The LN2 dewar shall be equipped with a ¼-inch vacuum seal-off valve for dewar evacuation by a turbomolecular pump.

2.4.1.3 The LN2 dewar shall be integrated with the camera body to provide ≥ 4 hour LN2 hold time while the camera is pointed in the zenith direction and tilted within $\pm 30^\circ$ of the vertical axis (perpendicular to the ground plane) while rotated $\pm 180^\circ$ about the optical axis, as shown in Figure 3.

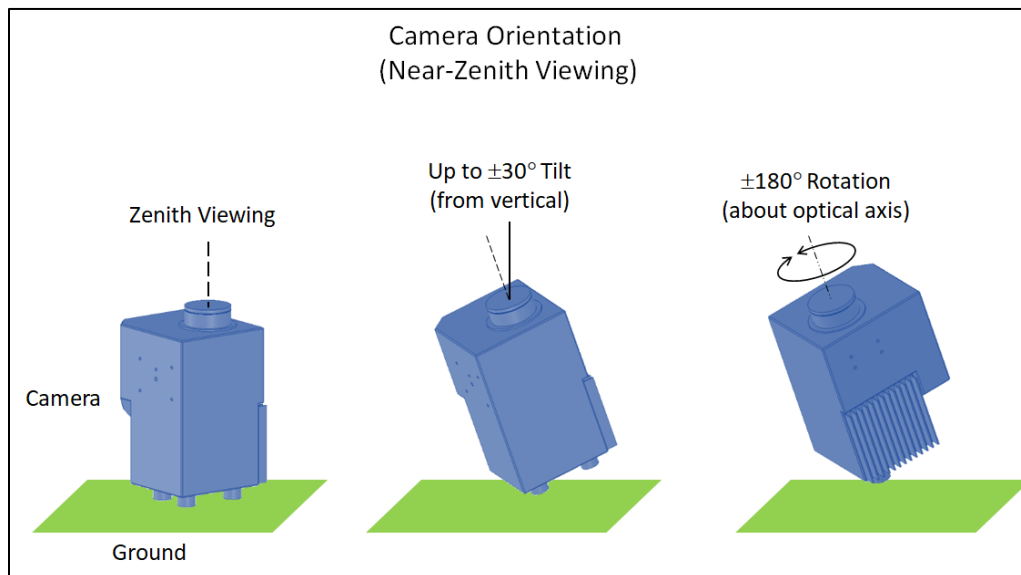


Figure 3 – LN2MW Camera Viewing Orientation

- 2.4.2 The integrated LN2MW Camera assembly including the LN2 dewar, camera electronics and filter wheel (but excluding the lens) shall be contained within the physical envelope defined in Figure 4.

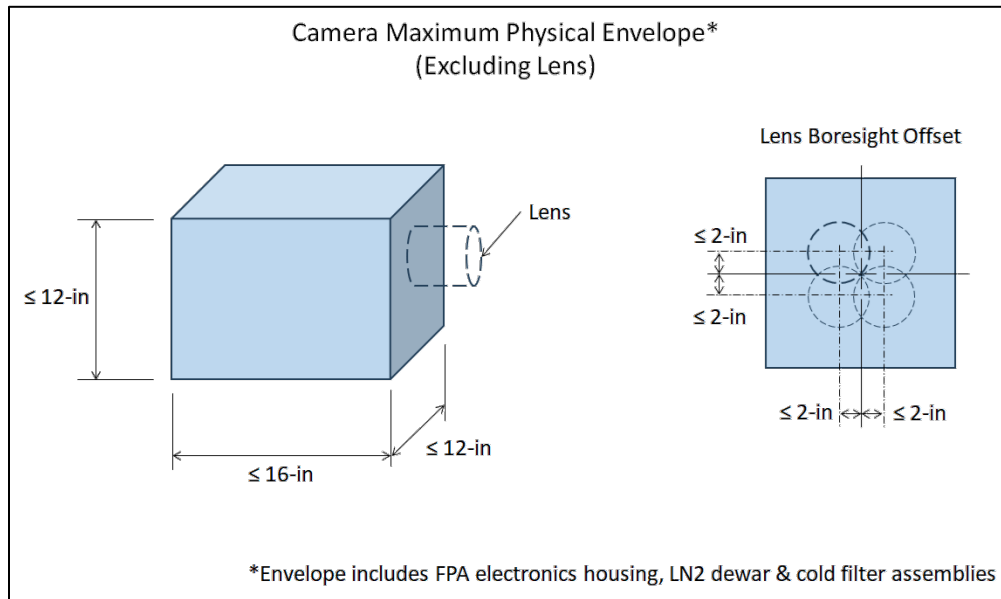


Figure 4 – LN2MW Camera Physical Envelope

- 2.4.3 Maximum camera weight (excluding lens): ≤ 10 lbs.
- 2.4.4 The LN2MW Camera shall include a minimum of two ¼-20 threaded holes for supporting and fastening the camera assembly to external mechanical hardware.

2.5 Software Control and Operation

- 2.5.1 The LN2MW Camera shall interface with the WinIRC software application for live camera video display, digital image recording and camera control from a host computer.
- 2.5.1.1 Single or multi-integration time (super-framing mode) selection.
 - 2.5.1.2 Discrete filter wheel position adjustment.
 - 2.5.1.3 NUC data creation and uploading to camera memory.
 - 2.5.1.4 Camera non-volatile memory reset to the factory default configuration for user-modifiable settings.
 - 2.5.1.5 Microsoft Windows 11 operating system operation (64-bit processor).
- 2.5.2 OPTIONAL: Software developer's kit (SDK)
- 2.5.2.1 Sample code in Visual Studio C++ for the user to develop custom software for communicating, configuring, and operating the camera from a host computer through the camera's Camera Link and Gigabit Ethernet interfaces.

2.5.2.2 Microsoft Windows 11 operating system operation

2.5.2.3 64-bit dynamic link library (DLL).

2.6 Environmental Compliance

2.6.1 Operating temperature: -40°C to +55°C.

2.6.2 Storage temperature: -55°C to +80°C.

2.6.3 Relative humidity: $\leq 95\%$ (non-condensing atmosphere).

2.7 Shipping/Storage Case

Hard-shell shipping/storage case for containing and transporting camera with lens equipment.

Appendix A

Appendix A: Optical Scattering Criteria

Background

Internal scattering by a thermal camera's optical system (including the focusing optics) can measurably affect the camera's FPA output with respect to the size of a subtended emission source imaged against a relatively uniform background scene. In extreme cases, the scattering phenomena can suppress the FPA's available dynamic range by effectively reducing the contrast of relatively hot or cold objects captured by the camera.

Definitions

Referring to Figure A-1, consider two extended area emission sources encompassed by regions, A_1 and A_2 , individually imaged by the camera from a fixed distance, where:

- A_1 contains nominally 10% of the FPA central pixels (significantly larger than the diffraction limited blur area).
- A_2 contains nominally 80% of the FPA central pixels.
- Assume the emission sources for A_1 and A_2 have identical emissivity and radiate diffusely.

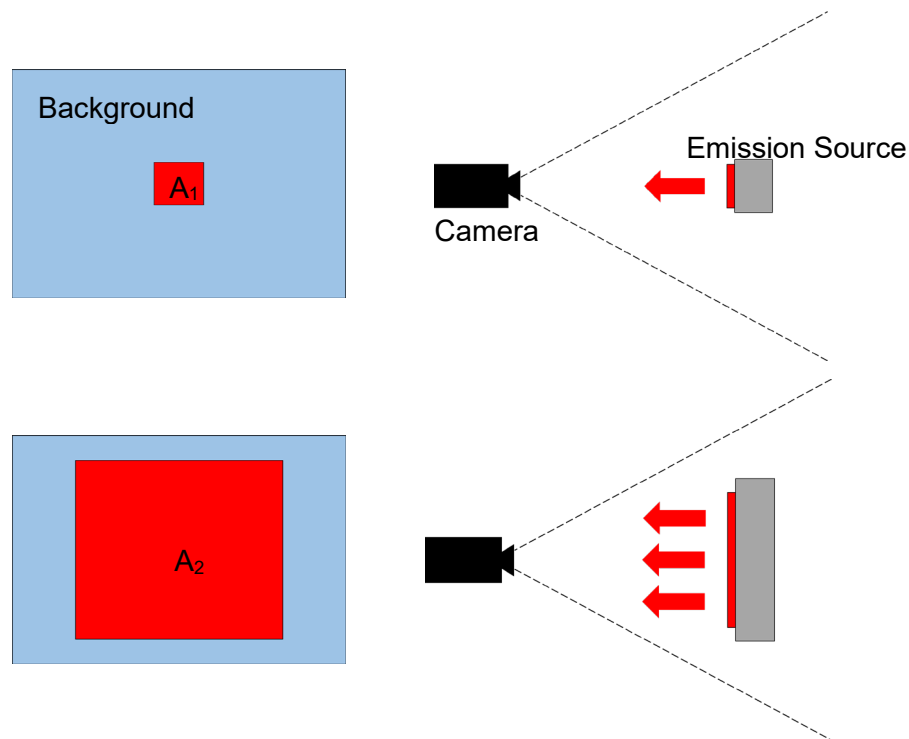


Figure A-1 – Emission Sources Imaged by the Camera

Methodology

The size of source sensitivity metric, S , is defined by: $S = \Delta X_1 / \Delta X_2$

By definition:

- $\Delta X_1 = X_1 @ T_S - X_1 @ T_B$
- $\Delta X_2 = X_2 @ T_S - X_2 @ T_B$
- X_1 and X_2 are the FPA pixel area averaged digital (count) values for respective regions A_1 and A_2 . The pixel areas exclude pixels in the blur region closest to a region's outer boundary
- T_S and T_B are the respective source and background temperatures (assumed to be spatially uniform), and are chosen such that ΔX_1 and ΔX_2 represent no less than a 50% change in detector well fill.
- T_S may be higher or lower than T_B

As illustrated by Figure A-2, S is derived from the slope of the curve generated from the collection and processing of camera imagery containing pixel regions A_1 and A_2 at source temperatures of T_S and T_B .

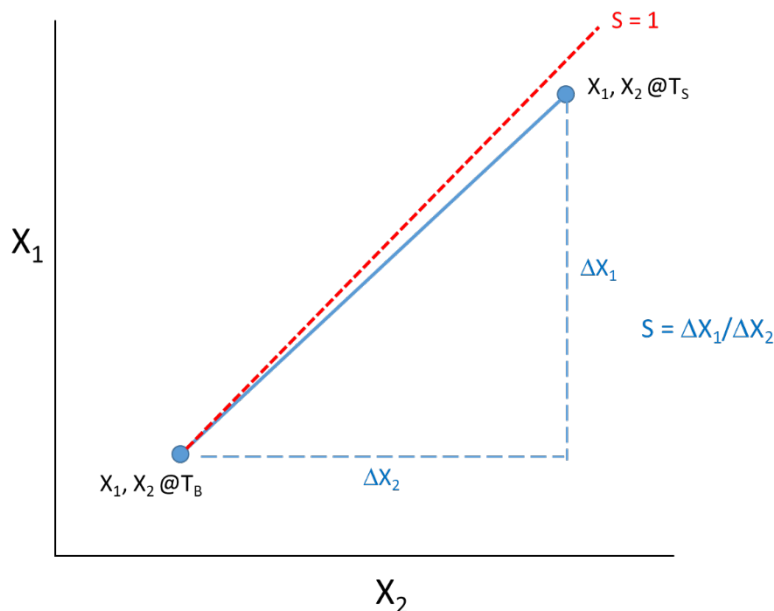


Figure A-2 – Determination of Optical Scatter Sensitivity, S

Summary

A camera system exhibiting no internal optical scattering of source emission rays between the FPA and focusing lens entrance aperture would ideally yield an S -value of unity (depicted by the red dash line in the figure). Otherwise, scattering occurring between the FPA and focusing lens will be prone to producing values of $S < 1$. Therefore, S can be used as a quantitative metric for establishing acceptable levels of internal optical scattering exhibited by a camera system.

Appendix B: Sample Certificate of Volatility

Date: _____

Certificate of Volatility				
Model:	Part Number:	Manufacturer:		
		Street Address:		
		City:	State:	Zip Code:
Volatile Memory				
Does the item contain volatile memory (i.e., memory whose contents are lost when power is removed)? ☐ Yes ☐ No				
If the answer is 'Yes', please provide the following information for each type (use additional sheets if required):				
Type (SRAM, DRAM, etc.):	Size:	User Modifiable: ☐ Yes ☐ No	Function:	Process to Sanitize:
Type (SRAM, DRAM, etc.):	Size:	User Modifiable: ☐ Yes ☐ No	Function:	Process to Sanitize:
Type (SRAM, DRAM, etc.):	Size:	User Modifiable: ☐ Yes ☐ No	Function:	Process to Sanitize:
Non-Volatile Memory				
Does the item contain non-volatile memory (i.e., memory whose contents are retained when power is removed)? ☐ Yes ☐ No				
If the answer is 'Yes', please provide the following information for each type (use additional sheets if required):				
Type (NVRAM, Flash, EEPROM, etc.):	Size:	User Modifiable: ☐ Yes ☐ No	Function:	Process to Sanitize:
Type (NVRAM, Flash, EEPROM, etc.):	Size:	User Modifiable: ☐ Yes ☐ No	Function:	Process to Sanitize:
Type (NVRAM, Flash, EEPROM, etc.):	Size:	User Modifiable: ☐ Yes ☐ No	Function:	Process to Sanitize:
Media				
Does the item contain media storage capability (i.e., removable or non-removable disk drives, tape drives, memory cards, etc.)? ☐ Yes ☐ No				
If the answer is 'Yes', please provide the following information for each type (use additional sheets if required):				
Type (Disk, Tape, etc.):	Size:	User Modifiable: ☐ Yes ☐ No	Function:	Process to Sanitize:
Removable: ☐ Yes ☐ No				
Type (Disk, Tape, etc.):	Size:	User Modifiable: ☐ Yes ☐ No	Function:	Process to Sanitize:
Removable: ☐ Yes ☐ No				
Type (Disk, Tape, etc.):	Size:	User Modifiable: ☐ Yes ☐ No	Function:	Process to Sanitize:
Removable: ☐ Yes ☐ No				
Additional Information:				
Manufacturer Representative Information				
Name:	Title:	Office Phone:	Fax/Email:	